FreeTTSVoice / IndianTTS: Project Status Report

Date: 17 October 2026
Status: Live Development & Phased Build
Stack: Next.js 15 (App Router) • Supabase (PostgreSQL) • Google Cloud TTS • Cloudflare Turnstile

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1. Executive Summary & Phasing Status

Phase	Scope	Status	Progress
`[NOW]` **Batch**	DB Consolidation, Reason Logging, Emoji Stripper, Turnstile, Real Limits	🟢 **COMPLETED & LIVE**	**100% (5/5)**
`[NEXT]` **Batch**	History Page, Saved Presets, Customer No (`#10000000+`), 10K Limit, WaveNet	🟢 **COMPLETED & LIVE**	**100% (6/6)**
`[SOON]` **Batch**	Junk Popup (R5), Feedback Bonus (R6), Runtime Copy (R8)	⏸️ **ON HOLD** *(Skipped by user)*	**0% (0/3)**
`[AFTER]` **Batch**	Razorpay Pro Packs, Multi-tier Upsell Modal, Voice Multipliers	⏳ **PENDING**	**0% (0/2)**

📌 Update Log: What Was Pending & Is Now Completed (Live)

The following 6 items were pending in the roadmap and have now been fully developed, tested, and shipped to live:

1. **R16 — Generation History Page (`/history`):**
   - *Previous Status:* Pending
   - *Current Status:* 🟢 **COMPLETED & LIVE**
   - *What was done:* Created app/history/page.tsx with inline audio playback, retention badge, and MP3 download; built backend app/api/history/route.ts querying `v_request_full`; added History link to Header.
2. **R14 — Saved Voice Presets (`tt_user_template`):**
   - *Previous Status:* Pending
   - *Current Status:* 🟢 **COMPLETED & LIVE**
   - *What was done:* Created app/api/templates/route.ts (CRUD); added "🚀 Save Preset" modal and "⭐ Presets" 1-click dropdown in components/tts/VoiceControls.tsx.
3. **R17 — Universal Customer Number (`seq_customer_no`):**
   - *Previous Status:* Pending

- *Current Status:* 🟢 **COMPLETED & LIVE**
- *What was done:* Auto-assign sequence starting from `10000000` via PostgreSQL sequence; displayed as `#10000001` badge on app/account/page.tsx ().

4. **R12 — Logged-in 10,000 Chars Limit + WaveNet:**
   - *Previous Status:* Pending
   - *Current Status:* 🟢 **COMPLETED & LIVE**
   - *What was done:* Synchronized free tier entitlements in lib/limits.ts () to 10,000 chars/req and 20,000 daily cap, unlocking WaveNet voices.
5. **R13 — Tier Retention Policy (Anon 30m / Free 7d):**
   - *Previous Status:* Pending
   - *Current Status:* 🟢 **COMPLETED & LIVE**
   - *What was done:* Built retention calculation into storage URLs and History UI with automatic expiration indicators.
6. **R11 — Protected Routes Middleware:**
   - *Previous Status:* Pending
   - *Current Status:* 🟢 **COMPLETED & LIVE**
   - *What was done:* Added `/history` and `/account` route protection in middleware.ts ().

2. Completed Requirements Matrix

🟢 Batch 1: `[NOW]` Core Foundation

Req ID	Requirement	Source / Config	Status	Implementation Details
R1	**Real-Time Limit Sync**	`pricing_plans.req_cap`	✅ **DONE**	Character counter in `components/tts/ScriptEditor.tsx` matches real backend validation ceiling.
R3	**Failure Reason & Excerpt Logging**	Always On	✅ **DONE**	`lib/usage.ts` captures `reason_code`, `error_message`, and 200-char `content_excerpt` in `tts_input_header`.
R4	**Smart Text Sanitizer**	`strip_emojis`	✅ **DONE**	`lib/tts/textSanitizer.ts` strips emojis & junk symbols before Google Cloud synthesis; punctuation (`. , ! ? - । ॥`) is strictly preserved.
R7 / R10	**Single Input Pipeline & Master View**	`v_request_full`	✅ **DONE**	Consolidated into `tts_input_header` + `tts_input_item`. Unused table parked as `tz_tts_inputs`. Master view `v_request_full` created.
R9	**Turnstile Bot Guard**	`turnstile_enabled`	✅ **DONE**	Cloudflare Turnstile token validation on TTS synthesis action only (not page load).

🟢 Batch 2: `[NEXT]` User Lifecycle & Login Perks

Req ID	Requirement	Source / Config	Status	Implementation Details
R11	**Server Session Auth**	Supabase SSR	✅ **DONE**	Cookie-based server sessions via `middleware.ts` and `app/auth/actions.ts`. Protected routes auto-redirect to `/login`.
R17	**Universal Customer Number**	`seq_customer_no`	✅ **DONE**	Pure bigint sequence starting from `10000000` auto-assigned to customers. UUID remains PK. Rendered as `#10000001` on `/account`.
R12	**10K Chars & WaveNet Unlocked**	`pricing_plans`	✅ **DONE**	Logged-in free tier unlocks **10,000 chars/req**, **20,000 daily cap**, and WaveNet voices. Governed by `lib/limits.ts`.
R13	**Tier Audio Retention Policy**	`tc_retention_policy`	✅ **DONE**	Anonymous guest audio kept for **30 minutes**; Free user audio kept for **7 days**.
R14	**Saved Voice Presets**	`tt_user_template`	✅ **DONE**	Built `app/api/templates/route.ts` and added "🚀 **Save Preset**" modal and "⭐ **Presets**" dropdown in `components/tts/VoiceControls.tsx`.
R16	**Generation History Page**	`/history`	✅ **DONE**	Built `app/history/page.tsx` + `app/api/history/route.ts` with inline audio playback, retention badges ("Kept for 7 days"), and MP3 download.

🟢 Additional Completed Core Features

- **Account Dashboard (`app/account/page.tsx`):** Live credit balance, active plan, purchase date, calculated expiry, billing history table, and username extraction (`ashutoshtiwari9453@...` → `@ashutoshtiwari`).
- **Navbar Redesign (`components/layout/Header.tsx`):** Removed confusing credit badges; added `@username` profile pill and "🎙️ **History**" navigation links for both desktop and mobile drawer.
- **Master Whitelabel Configuration (`site.ts` & `config/site.ts`):** Centralized website name, domain, legal, social, audio defaults, and footer text.

3. Pending Requirements (Work Remaining)

Phase 1: `[AFTER login]` — Monetization & Pro Tier

Req ID	Feature	Priority	Implementation Plan
R15	**Razorpay Pro Packs & Voice Multipliers**	High	Razorpay order creation and webhook verification. Credit consumption based on voice multipliers (Standard 1.0x, WaveNet 1.5x, Neural2 2.0x). Persistent "Upgrade to Pro" banner.
R2	**Multi-tier Character Upsell Modal**	High	When guest users enter > 1,000 characters, trigger an upsell modal prompting free signup to unlock 10,000 characters + WaveNet voices.

Phase 2: `[SOON]` — Quality & Feedback (Currently On Hold)

Req ID	Feature	Priority	Implementation Plan
R5	**Junk Text Detection Popup**	Medium	Check text against heuristic thresholds (`junk_min_letters = 2` , `junk_max_repeat = 20`). If blatant junk detected, show confirmation modal: "*Your text appears repetitive. Generate anyway?*"
R6	**Feedback Form & Bonus Reward**	Medium	Studio modal allowing users to submit feedback into `tt_feedback` . First submission awards **10,000 bonus characters**.
R8	**Runtime Editable Site Copy**	Low	Query `tc_site_copy` for dynamic homepage copy without re-deployments, with automatic fallback to `site.ts` .

4. Test Cases Verification Table

Test ID	Test Scenario	Expected Outcome	Live Verification Status
TC-R1	Anonymous user loads Studio	Counter reads `/1,000`; typing 1,001 characters shows validation warning	✅ **PASSED**
TC-R3	Generation failure / over-cap	`tts_input_header` has `reason_code = text_too_long`, `error_message`, and `content_excerpt`	✅ **PASSED**
TC-R4	Input with emojis: `"Hello 😀 World 🚀!"`	Google TTS synthesizes `"Hello World!"` with clean pauses; original stored in DB	✅ **PASSED**
TC-R9	Generate action without Turnstile	Request rejected with `turnstile_failed`; zero Google TTS provider cost incurred	✅ **PASSED**
TC-R11	User logs in and visits protected routes	Session persists across tabs; unauthorized users redirected to `/login?next=...`	✅ **PASSED**
TC-R12	Logged-in user loads Studio	Counter automatically expands to `/10,000`; WaveNet voices selectable	✅ **PASSED**
TC-R14	User saves preset `"Hindi Dev"`	Preset stored in `tt_user_template`; selecting from dropdown applies settings	✅ **PASSED**
TC-R16	User opens `/history` page	Shows only user's own generations with inline audio player and "Kept for 7 days" badge	✅ **PASSED**
TC-R17	Customer number assignment	New user receives sequential bigint ≥ 10000000; displayed on `/account` as `#10000001`	✅ **PASSED**

5. File Manifest & Key Locations

```
c:\Users\lenovo\Desktop\mine (1)\
├── app/
│   ├── account/page.tsx              <-- Account Dashboard (Customer #ID, Credits, Plans)
│   ├── history/page.tsx              <-- Generation History & Audio Player (R16)
│   ├── text-to-speech/page.tsx       <-- Main IndianTTS Studio
│   └── api/
│       ├── generate/route.ts         <-- TTS synthesis, Turnstile & Sanitizer
│       ├── history/route.ts          <-- User history endpoint (v_request_full)
│       ├── templates/route.ts        <-- Saved voice presets API (tt_user_template)
│       ├── user/credits/route.ts     <-- User balance, tier, and customer_no
│       ├── user/subscription/route.ts<-- Invoices and plan metadata
```

```
│       └── download/route.ts          <-- Branded MP3 download proxy
├── components/
│   ├── tts/
│   │   ├── ScriptEditor.tsx           <-- Character counter synced with backend
│   │   └── VoiceControls.tsx          <-- Language selector + Preset bookmarks
│   └── layout/
│       ├── Header.tsx                 <-- Navbar with @username & History link
│       └── Footer.tsx                 <-- Whitelabel footer using site.ts
├── lib/
│   ├── limits.ts                      <-- Tier definitions (Anon 1K, Free 10K, Paid 10K+)
│   ├── credits.ts                     <-- Balance checks, customer_no sequence logic
│   ├── user.ts                        <-- Username sanitization (ashutoshtiwari9453 ->
ashutoshtiwari)
│   └── tts/
│       ├── textSanitizer.ts           <-- Emoji & symbol stripper
│       └── generationService.ts       <-- Single input orchestration & audit logging
└── config/
    └── site.ts                        <-- Master single-source whitelabel configuration
```

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6. Build & System Health

- **TypeScript Health:** `tsc --noEmit` exited with code `0` (Zero compilation errors).
- **Database Status:** Supabase PostgreSQL migrations fully active (`tts_input_header` , `tt_user_template` , `seq_customer_no` , `v_request_full`).
- **Dev Server:** Active and operational on `http://localhost:3000` .